

Temple University IR Project

Weekly narrative (auto-generated)

May 3, 2026

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System overview and challenges

End-to-end flow

1. **Crawl** (`src/crawl_temple.py`): BFS on `temple.edu`, stream-append JSONL, resume by scanning existing docno and URLs. Two queues prefer unseen URLs before re-fetching known pages (expansion-only).
2. **Dedupe** (`src/dedupe_corpus.py`):
`first row per docno` → `data/raw/corpus_deduped.jsonl`.
3. **Index** (`src/build_index.py`): PyTerrier `IterDictIndexer`, meta `docno`, text field `text`.
4. **Analysis** (`src/collection_analysis.py`): vocabulary, Zipf, lengths, stopwords, word cloud → `outputs/analysis/`.
5. **Queries / pooling**: templates and `queries.tsv`; pool from BM25, TF-IDF, QL.
6. **Bootstrap qrels**: rank-1 BM25 labeled relevant (temporary metrics only).

7. **Eval, sensitivity, optimization:** PyTerrier experiments → `outputs/eval/`, `outputs/optimization/`.
8. **Reports:** `src/update_reports.py` refreshes Markdown; `src/export_latex_reports.py` emits this PDF bundle.

Snapshot numbers (from this export)

- Raw JSONL lines: **71876** (includes duplicate lines from restarts).
- Deduped documents: **42217**.
- Collection summary (first CSV row): docs 42217.0, vocab 214003.0, tokens 37491119.0, avg length 888.0573939408296.
- BM25 nDCG@10 (aggregate file, bootstrap qrels): **0.884123967142861**.

Main challenges encountered

- **Resume and duplicates:** repeated crawl runs appended lines; some URLs mapped to multiple docno values over time. Dedupe-by-docno plus index-time `drop_duplicates` kept Terrier stable.
- **Frontier / scheduling:** treating “already saved” URLs as non-expandable starved BFS; fixed by always enqueueing unvisited links and then splitting *new* vs *expand* queues so new URLs are tried first.
- **Scale vs site reality:** many hosts return HTTP 403 to datacenter or script clients; tqdm only moves on *new* docs so long runs looked idle while HTTP work continued—postfix counters (`fetched`, queue sizes) were added for visibility.
- **PyTerrier/Java:** JVM init, absolute index paths, `text_attrs`, unique docno constraints required several integration passes.
- **Evaluation wiring:** PyTerrier Experiment metric tables differ in wide vs aggregate layouts; bootstrap qrels are not a substitute for human `qrels.tsv`.

1 Week 1: Data collection

Crawl `temple.edu` with politeness delay; stream to `data/raw/corpus.jsonl`. Resume scans the file once for max docno suffix and known URLs (memory-safe). Deduped count for this snapshot: **42217** unique documents (from **71876** raw lines).

2 Week 2: Indexing and collection analysis

Build PyTerrier index under `index/terrier`. Run collection statistics: file-type table, vocabulary and token counts, Zipf log-log plot, document-length distribution, stopword table, word cloud—all under `outputs/analysis/`. Crawl is HTML-only; PDF/DOCX would need a separate ingest.

3 Week 3: Collaborative query construction

Fifteen structured queries in `data/processed/query_templates.tsv` with intent metadata; `queries.tsv` feeds PyTerrier runs.

4 Week 4: Pooling and relevance

TREC-style pool (BM25 + TF-IDF + QL, depth 10) → `pooled_candidates.tsv`. Official `qrels.tsv` remains for human labeling; bootstrap `qrels` exist only to exercise metrics.

5 Week 5: Baseline retrieval

BM25, TF-IDF, and Query Likelihood evaluated with bootstrap `qrels` until human judgments exist. See `outputs/eval/aggregate_metrics.csv` and per-query exports.

6 Week 6: BM25 sensitivity

Grid over k_1 and b with nDCG@10 curves in `outputs/eval/bm25_sensitivity_curves.png`.

7 Week 7: Optimization experiment

Compare conjunctive vs disjunctive BM25 variants; record latency, rows returned, and overlap@10 in `outputs/optimization/optimization_results.csv`.

8 Week 8: Modern extension

Not implemented in-repo (Elasticsearch or LLM rerank). Reserved for future work.

9 Week 9: Error analysis

Partial: template in `src/error_analysis_template.md`; use low nDCG@10 rows from `per_query_metrics.csv` with manual pool inspection.

10 Week 10: Final integration

Consolidated Markdown report at `reports/final_report.md`; rubric checklist at `reports/DELIVERABLES.md`. This LaTeX bundle mirrors the same timeline.